

WHAT APP?

KEY IDEA

The Metric System (SI)

The **International System of Units (SI)** is the standard system of measurement used worldwide.

Metric units are related by powers of 10 — that's what makes converting so simple:



Larger → Smaller Unit

MULTIPLY by the power of 10

(move the decimal point RIGHT)



Smaller → Larger Unit

DIVIDE by the power of 10

(move the decimal point LEFT)

Example 1

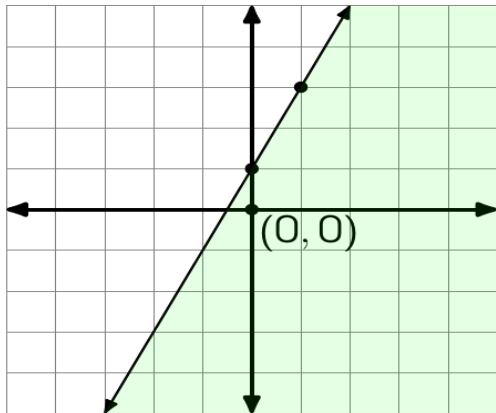
Step 4: If the inequality is satisfied, shade the region containing that point. If not, shade the other region.

Original inequality $y \leq 2x + 1$

Plane divider $y = 2x + 1$

$m = 2$

$b = 1$



Let $x = 0, y = 0$

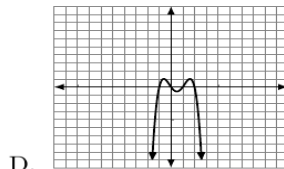
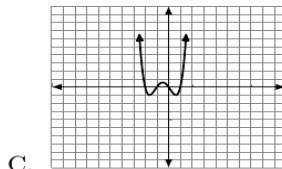
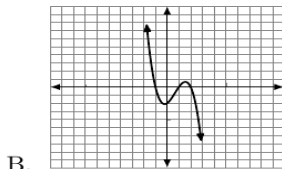
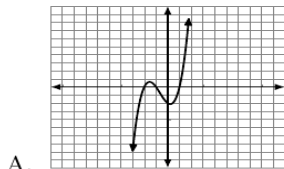
$$y \leq 2x + 1$$

$$0 \leq 2(0) + 1$$

$$0 \leq 0 + 1$$

$$0 \leq 1 \quad \checkmark$$

16. Which of the following could be the graph of the polynomial function $y = -x^3 + 2x^2 + x - 2$?



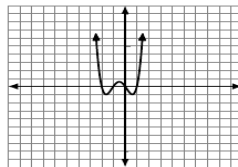
17. What are the end behaviors of the graph of $f(x) = -2x + x^3 + 3x^5 - 4$?

- A. rises to the left and falls to the right
B. falls to the left and rises to the right

- C. rises to both directions
D. falls to both directions

18. Which polynomial function in factored form represents the given graph?

- A. $y = (x + 2)(x + 1)(x - 1)$
B. $y = (x - 2)(x + 1)(x - 1)$
C. $y = x(x + 2)(x + 1)(x - 1)$
D. $y = x(x - 2)(x + 1)(x - 1)$



19. An arc with a measure equal to one-half the circumference of a circle is called:

- A. Intercepted arc B. Major arc C. Minor arc D. Semicircle

20. If $m\widehat{AC} = 40^\circ$ and $m\widehat{BD} = 80^\circ$, find $m\angle ATC$.



Mathematics 8

Module 6 – Linear Equations

Introduction: This module was designed and written with you in mind. It is specially made for you to gain a thorough understanding on several concepts on linear equations. Specifically, you will be able to write the linear equation $Ax + By = C$ in the form $y = mx + b$ and vice versa, graph a linear equation given (a) any two points; (b) the x- and y- intercepts; (c) the slope and the y-intercept; and (d) the slope and a point on the line, and describe the graph of a linear equation in terms of its intercepts and slope.

Lesson 1

Forms of Linear Equation



What To Know

This module is designed for grade 8 students like you, so that you will have a better understanding on the concepts of linear equations. Specifically, you are expected to:

1. write the linear equation $Ax + By = C$ in the form $y = mx + b$ and determine the values of m and b ;
2. write the linear equation $y = mx + b$ in the form $Ax + By = C$ and determine the

- graph a linear equation given the slope and the y-intercept;
- graph a linear equation given the slope and a point on the line.

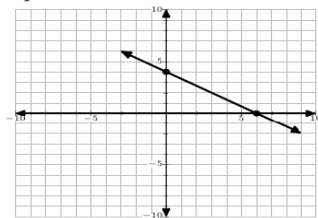


Pre-Assessment

Answer the following questions. Choose the letter that corresponds to the correct answer.

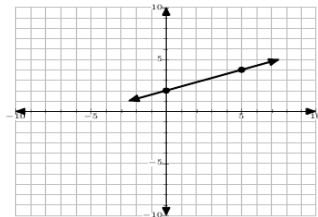
1. Which of the following information may be used to plot the graph at the right?

- A. $a = 4$ and $b = 6$
- B. $m = 4$ and $b = 6$
- C. $a = 6$ and $b = 4$
- D. $m = 6$ and $b = 4$



2. The graph at the right can be described by which of the following information?

- A. $a = 2$ and $m = \frac{2}{5}$
- B. $b = 2$ and $m = \frac{2}{5}$
- C. $a = 2$ and $b = \frac{2}{5}$
- D. $b = 2$ and $a = \frac{2}{5}$



**“Math Hugot”:
An Intervention for Improving
Attitudes toward Mathematics Among
Grade 10 Students of
Sauyo High School,
S.Y. 2019–2020**

Jonathan R. Bacolod, LPT

Researcher

Quiz 3.6: Proving the Congruence of Triangles

Multiple Choice: Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.

1. Which theorem states that if the legs of one right triangle are congruent to the legs of another right triangle, then the triangles are congruent?

A. HA Congruence Theorem

C. LA Congruence Theorem

B. HL Congruence Theorem

D. LL Congruence Theorem

2. "If two angles and a non-included side of one triangle are congruent to the corresponding two angles and a non-included side of another triangle, then the triangles are congruent." This is stated in:

A. AAS Congruence Theorem

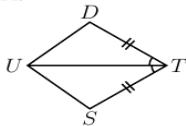
C. HL Congruence Theorem

B. LL Congruence Theorem

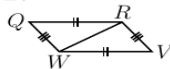
D. LA Congruence Theorem

3. Which of the following pairs of triangles are congruent and can be proved by ASA Postulate?

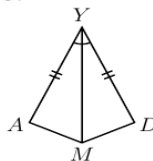
A.



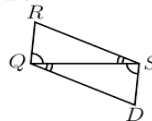
B.



C.



D.



4. Which of the following pairs of triangles are congruent and can be proved by HL Theorem?

Worksheet 1.5.3: Slope of a Line

A. True or False

Write True if the statement is true or False if it is false. One point each.

1. The slope of the line described by the equation $y = -3x$ is -3 .
2. The slope of the line described by the equation $y = 4x - 1$ is -1 .
3. The slope of the line described by the equation $y = \frac{3}{2}x + 2$ is 3 .
4. The slope of the line described by the equation $y = -\frac{5}{4}x - 3$ is $-\frac{5}{4}$.
5. The slope of the line described by the equation $5x + y = 3$ is 5 .

B. Finding the Slope Given Two Points

Find the slope of the line that passes through each pair of points. Choose the answer from the box. Some answers may be repeated. Write the letter only. One point each.

a. -4

e. $\frac{3}{4}$

b. -1

f. $\frac{3}{5}$

c. -10

g. $-\frac{3}{4}$

d. $\frac{15}{8}$

h. 1

6. $(4, 3)$ and $(2, 1)$

11. $(9, -7)$ and $(3, -1)$

7. $(-3, 2)$ and $(-5, 4)$

12. $(5, -4)$ and $(9, -7)$



	School	SAUYO HIGH SCHOOL	Grade Level	8
	Teacher	JONATHAN R. BACOLOD	Learning Area	Mathematics
	Teaching Date and Time	Nov 17, 2025 8 – Oman and Qatar 12:30 PM – 6:50 PM	Quarter	3rd

I. OBJECTIVES	
A. Content Standard	The learner demonstrates knowledge and understanding of linear equations in two variables and their graphs.
B. Performance Standard	The learner is able to graph linear equations in two variables. (NA)
C. Content Domain	Number and Algebra (NA)
D. Most Essential Learning Competencies/Objectives	find the equation of a line given the slope and a point
II. CONTENT	Finding the Equation of a Line Given the Slope and a Point
III. LEARNING RESOURCES	
A. References	
Teacher's Guide	Teacher-made Learning Worksheet
Learner's Materials	Activity Sheets
B. Other Learning Resources	Terpanel, Printed Materials, Activity Sheets

Republic of the Philippines
 Department of Education
 National Capital Region
SCHOOLS DIVISION OFFICE QUEZON CITY
SAUYO HIGH SCHOOL
 Pantabangan St., NIA Village, Sauyo, Quezon City

MATATAG K to 10 Curriculum Weekly Lesson Exemplar	School	SAUYO HIGH SCHOOL	Grade Level	8
	Name of Teacher	JONATHAN R. BACOLOD	Learning Area	Mathematics
	Teaching Dates and Time	December 8–12, 2025, 12:30 PM – 6:50 PM	Quarter	Quarter 3, Week 6

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
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I. CURRICULUM CONTENT, STANDARDS, AND LESSON COMPETENCIES

A. Content Standards	The learners demonstrate knowledge and understanding of: 1. systems of linear equations in two variables				
B. Performance Standards	By the end of the quarter, the learners are able to: 1. solve a system of linear equations graphically and algebraically				
C. Learning Competencies	By the end of the lesson, the learners: 1. define and illustrate a system of linear equations in two variables 2. solve a system of linear equations (with integer solutions) by graphing 3. classify the types of systems of linear equations based on the number of solutions				

	The learner should be able to: a. Recall the concepts	The learner should be able to: a. Outline and estimate	The learner should be able to: a. Solve and analyze the various	The learner should be able to: a. Cite and analyze the various components involved in classifying types of	The learner should be able to: a. Recall the concepts
--	--	---	--	---	--

Inscribed Angles and Intercepted Arcs

Inscribed angle: an angle whose vertex lies on the circle and whose sides are chords of a circle

The measure of an inscribed angle is **half** the measure of its intercepted arc.

In a circle, if two inscribed angles intercept the same arc or congruent arcs, then the angles are congruent.

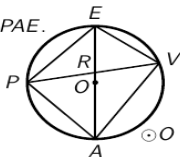
An angle inscribed in a semicircle is a right angle and therefore the measure is equal to 90° .

If a quadrilateral is inscribed in a circle, then its opposite angles are supplementary.

Practice Exercises

A. Refer to $\odot O$ to answer the following.

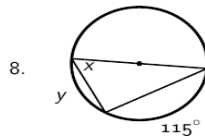
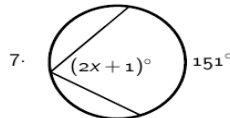
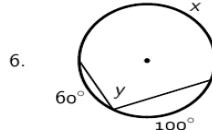
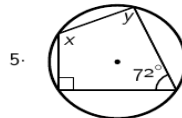
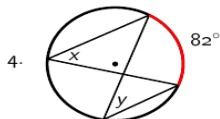
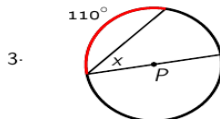
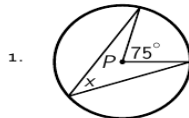
1. Name the angle that intercept $\widehat{AP}$.
2. Name the angles that intercept $\widehat{EV}$.
3. Name the arc that is intercepted by $\angle PAE$.
4. Name the arc that is intercepted by $\angle EVP$.
5. If $m\angle PEA = 48^\circ$, then $m\widehat{AP} = \underline{\hspace{2cm}}$ and $m\angle AVP = \underline{\hspace{2cm}}$.
6. $m\angle EPA = \underline{\hspace{2cm}}$
7. $m\angle EVP + m\angle PVA = \underline{\hspace{2cm}}$
8. If $m\angle VEP = 100^\circ$, then $m\angle PAV = \underline{\hspace{2cm}}$.



B. Given $\odot S, \overline{AR} \cong \overline{RO} \cong \overline{OS} \cong \overline{SA}$, $m\angle AMR = 3x + 20$ and

Problem Set

A. Use the given figures to find the value of x and y .



B. $\triangle GOA$ is inscribed in $\odot L$. If $m\angle OGA = 75^\circ$ and $m\widehat{AG} = 160^\circ$, find:

1. $m\widehat{OA}$

Grade 8 Math Handouts



3RD QUARTER

Lesson 3.1: Solving Linear Equations in One Variable

 [Share Link](#)

Lesson 3.1: Solving Linear Equations in One Variable

TEEMathPro: Math Teaching Beautified

Linear equation: an equation where the variable is raised to the power of one, and the equation represents a straight line when graphed.

VOLUME OF A CYLINDER: FROM CIRCLE AREA TO 3D SPACE

Discovering $V = \pi r^2 h$ through Inductive Stacking & Pattern Recognition

1. What is Inductive Reasoning?

Inductive Reasoning means observing specific patterns and building up to a general formula!

Concept Bridge: Flat 2D Circle (Coin/Pancake) → Stacking h layers → 3D Solid Cylinder!

2. Graphic Organizer: The Pancake Stack

A circular coin/pancake has a base radius r and base area $A = \pi r^2$.

Inductive Stacking Process



 **GRADE 7 MEMORY HOOK (MNEMONIC)**

"PIES ARE SQUARED HIGH!" 

Picture a tall stack of delicious circular pies piled up **HIGH!**

PIE (π) • ARE (r) • SQUARED (2) • HIGH (h) → $V = \pi r^2 h$

THE CYLINDER VOLUME FORMULA

$$V = \text{Base Area} \times \text{Height} = \pi r^2 h$$

Base Area (B) = Circle Area = πr^2



4. Worked Example: Soda Can Calculation

Problem: A soda can has a base radius $r = 3 \text{ cm}$ and height $h = 10$

Parallelism and Perpendicularity

Word List

A	T	T	F	C	J	B	A	P	S	R	Y	P	P	I	F	T	A	F	B
G	T	B	X	O	X	B	C	F	G	A	Y	E	L	S	B	V	L	C	H
I	M	A	T	N	G	D	J	K	L	N	Y	R	A	D	K	V	T	G	A
N	W	L	M	S	U	I	X	Y	O	O	M	P	M	U	N	X	E	E	N
T	K	T	P	E	N	S	E	D	U	F	Y	E	S	G	W	G	R	J	G
E	I	E	V	C	J	T	M	P	G	G	T	N	A	S	Z	Z	N	E	L
R	U	R	I	U	T	T	V	A	C	T	T	D	F	I	B	X	A	C	E
S	H	N	B	T	A	R	H	R	M	I	K	I	O	C	P	I	T	O	R
E	A	A	M	I	F	A	L	A	Y	C	S	C	C	T	B	Y	E	R	E
C	U	T	A	V	N	N	I	L	E	R	K	U	Q	R	I	M	I	R	L
T	P	E	T	E	O	S	N	L	Q	E	E	L	M	I	R	I	N	E	A
I	J	E	H	I	R	V	E	E	U	A	W	A	C	G	M	D	T	S	T
O	C	X	P	N	W	E	S	L	I	S	L	R	H	H	S	S	E	P	I

PARALLELLINES

PERPENDICULAR

TRANSVERSAL

ANGLERELATIONSHIP

CORRESPONDING

ALTERNATEINTERIOR

ALTERNATEEXTERIOR

CONSECUTIVEINTERIOR

RIGHTANGLE

LINESEGMENTS



Republic of the Philippines
Department of Education
National Capital Region
SCHOOLS DIVISION OFFICE OF QUEZON CITY
MALIGAYA HIGH SCHOOL
Ilang Ilang St., Maligaya Park Subd.
Pasong Putik, Nova., Quezon City

Name: _____
Section: _____

Date: _____
Score: _____

Third Summative Test in Mathematics 8
S.Y. 2023–2024

Multiple Choice: *Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.*

1. A point can be represented as:
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
2. Which of the following best represents a plane?
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
3. A structure that consists of defined and undefined terms, axioms or postulates, and theorems is called:
A. Direct proof B. Indirect proof C. Law of Syllogism D. Mathematical system
4. Two lines that are not parallel ____ intersect.
A. always B. sometimes C. maybe D. never
5. Evaluate the following statement: "Perpendicular lines form right angles."

Grade 8 Fourth Quarter Reviewer

Jonathan R. Bacolod, LPT

TeXMon^hPro: Math Teaching Beautified

March 10, 2025

Applying Theorems on Triangle Inequality

The Triangle Inequality Theorem states that the sum of the lengths of any two sides of a triangle must be greater than the length of the third side:

$$a + b > c, \quad a + c > b, \quad b + c > a$$

Basic Concepts of Probability

Probability measures the likelihood of an event occurring, expressed as:

$$P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

where $0 \leq P(A) \leq 1$.

Counting the Outcomes of Experiments

- Use the Fundamental Counting Principle: If one event has m outcomes and another event has n outcomes, the total number of outcomes is $m \times n$.
- Tree diagrams and tables help visualize possible outcomes.

Experimental and Theoretical Probability

- **Theoretical Probability:** Based on mathematical reasoning.
- **Experimental Probability:** Based on actual experiments.

Parallelism and Perpendicularity

- Parallel lines never intersect and have the same slope.
- Perpendicular lines intersect at 90° and their slopes are negative reciprocals.

Probability of an Event

Given an event A , its probability is calculated as:

$$P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

If A' is the complement of A :

$$P(A') = 1 - P(A)$$

Proving Inequalities in a Triangle

To prove triangle inequalities, use:

- Triangle Inequality Theorem.
- Exterior Angle Inequality Theorem.
- Hinge Theorem.

Proving Properties of Parallel Lines Cut by a Transversal

When two parallel lines are cut by a transversal:

- Corresponding angles are congruent.

$$4\sqrt{6} + 3\sqrt{6}$$

B^A

$$6\sqrt{5}$$

TPXMa[®]Pro Math Teaching Beautiful
Created by Jonathan R. Dwyer, LPT
Adding Radical Expressions

B^A + B^A

$$5\sqrt{7} + \sqrt{7}$$

L^A

$$3\sqrt{10} + 4\sqrt{10}$$

B^A

$$9\sqrt{2}$$

TPXMa[®]Pro Math Teaching Beautiful
Created by Jonathan R. Dwyer, LPT
Adding Radical Expressions

B^A + B^A

$$\sqrt{3} + 4\sqrt{3}$$

B^A

$$4\sqrt{6} + \sqrt{6}$$

B^A

$$3\sqrt{2}$$

TPXMa[®]Pro Math Teaching Beautiful
Created by Jonathan R. Dwyer, LPT
Adding Radical Expressions

B^A + B^A

$$4\sqrt{7} + 6\sqrt{7}$$

$$6\sqrt{5} + 2\sqrt{5}$$

B^A

$$5$$

TPXMa[®]Pro Math Teaching Beautiful
Created by Jonathan R. Dwyer, LPT
Adding Radical Expressions

B + B^A

$$\sqrt{8} + 3\sqrt{2}$$

$$2\sqrt{3} + \sqrt{12}$$

$$7\sqrt{2}$$

$$2 + \sqrt{4}$$

Factoring Trinomials Flashcards

by Jonathan R. Bacolod, LPT

Flashcard 1 of 10

$$n^2 - 2np + p^2$$

Type factored form, e.g. $(2x + 1)(x - 9)$

Previous

Next

Submit

Leader's Name:

Section:

Group Number:

[illegible]

OMR Scanner

Capture-on-demand · marker-based perspective correction ·
calibratable bubble grid

Live — start Auto-Scan.

Setup

The 3 things you set per test.

Section

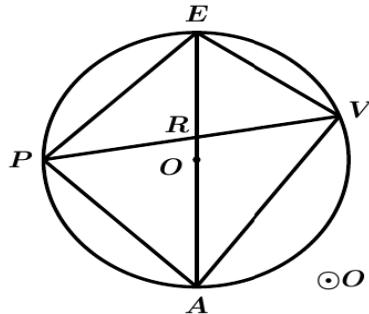
7-Malvar

Questions

50

Practice Exercises

A. Refer to the figure on the right to answer the following.



1. Name the angles that intercept $\widehat{AP}$.
2. Name the angles that intercept $\widehat{EV}$.
3. Name the arc that is intercepted by $\angle PAE$.
4. Name the arc that is intercepted by $\angle EVP$.
5. If $m\angle PEA = 48$, then $m\widehat{AP} = \underline{\hspace{1cm}}$ and $m\angle AVP = \underline{\hspace{1cm}}$.
6. $m\angle EPA = \underline{\hspace{1cm}}$
7. $m\angle EVP + m\angle PVA = \underline{\hspace{1cm}}$
8. If $m\angle VEP = 100$, then $m\angle PAV = \underline{\hspace{1cm}}$.



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How to Memorize Math Formulas

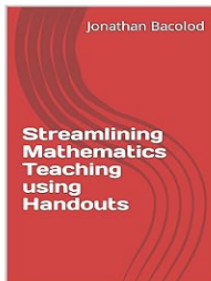


A Comprehensive Guide to Memorization and Retention for High School Teachers and Students

Jonathan R. Bacolod, LPT

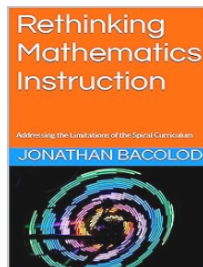
Quick look

How to Memorize Math Formulas: A Comprehensive Guide to Memorization...

Kindle Edition\$0⁰⁰ **kindle**unlimitedor **\$9.99** to buyOther formats: [Paperback](#)

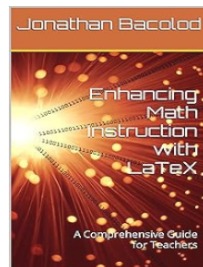
Quick look

Streamlining Mathematics Teaching using Handouts

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Quick look

Rethinking Mathematics Instruction: Addressing the Limitations of the...

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Quick look

Enhancing Math Instruction with LaTeX: A Comprehensive Guide for Teachers

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WHAT APP?



LARGE LANGUAGE MODELS

LLM

AI CHATBOT

CHAT

Why can't you
harness the
power of AI?

BASIC
PROMPT
ENGINEERING
FOR HIGH SCHOOL
MATH TEACHERS

By the end of the workshop,
every teacher leaves with at
least 3–5 AI-generated teaching
materials they can actually use.

TRY IT

Create a
worksheet
about fractions.

PROBLEM:
AI ×
PROMPT ✓

WHAT IS
PROMPT
ENGINEERING?

TALKING TO AI EFFECTIVELY

WHY
CARE?

SAVE
TIME



CREATE
BETTER
MATERIALS

4. A relation can be presented using the following ways EXCEPT:

- a. arrow diagram b. ordered pairs c. graph d. line

5. If range is the dependent variable, _____ is the independent variable.

- a. domain b. abscissa c. ordinate d. relation

6. A special type of relation, in which every element in the domain is mapped to exactly one element in the range, is called a _____.

- a. relation b. ordered pair c. ordinate d. relation

7. Identify which graph represents a function.



8. Which set of ordered pairs define a function?

- a. $\{(4,0),(4,1),(4,2)\}$ b. $\{(2,1),(2,3),(4,5)\}$ c. $\{(1,2),(3,2),(5,2)\}$ d. $\{(1,1),(-1,2),(-1,3)\}$

9. If every _____ line intersects the graph no more than once, the graph represents a function.

- a. horizontal b. vertical c. diagonal d. parallel

10. This notation can be used to define a function.

- a. Scientific notation b. formative notation c. function notation d. relation

11. The following are considered linear function EXCEPT:

- a. $f(x)=2x+3$ b. $y=2x+3$ c. $2x+3$ d. y

DIFFERENTIATE
INSTANTLY

TRY IT

Create 10
Math
problems.

Create 10 Grade 8 problems
on solving linear equations
involving variables on both
sides.

The “SCARE” Framework

SPECIFIC
CONTEXT
ASSIGN A ROLE
REFINE
EXAMPLES

PRINCIPLE #1

BE
SPECIFIC

What exactly
do I want?

VAGUE:
“Create a worksheet”

SPECIFIC:

“Create 10 quadratic
equation problems
for grade 9 students”

VAGUE:
“Plan a lesson on circles”

SPECIFIC:

“Create a 45-min lesson plan on arc length for grade 10, including a Ferris wheel task”

VAGUE:
“Make word problems”

SPECIFIC:

“Write 5 linear systems
word problems where the
answers are whole numbers”

VAGUE:

“Help a student solve x ”

SPECIFIC:

“Ask 3 Socratic questions to help a student isolate x in $3x + 5 = 20$ without telling them”

VAGUE:

“Make this trig task harder”

SPECIFIC:

“Rewrite these trig problems
for Honors level using only
exact radical forms”

VAGUE:
“Check this math work”

SPECIFIC:

“Find the sign error in this polynomial division and write a hint for the student”

PRINCIPLE #2

GIVE
CONTEXT

Who are
my learners?

What do they
already know?

What are they
struggling
with?

“My students
struggle with
FRACTIONS”

“They’re in
BASIC ALGEBRA”

“Reading level:
GRADE 7”

“They know basic
probability, but not
COMBINATIONS”

“My class is
obsessed with
MINECRAFT”

“I have no
tech and only
15 MINUTES”

PRINCIPLE #3

ASSIGN
A ROLE

Who should
AI act as?

ACT AS A
CURRICULUM
PLANNER

Republic of the Philippines
 Department of Education
 National Capital Region
SCHOOLS DIVISION OFFICE QUEZON CITY
SAUYO HIGH SCHOOL
 Pantabangan St., NIA Village, Sauyo, Quezon City

MATATAG K to 10 Curriculum Weekly Lesson Exemplar	School	SAUYO HIGH SCHOOL	Grade Level	8
	Name of Teacher	JONATHAN R. BACOLOD	Learning Area	Mathematics
	Teaching Dates and Time	December 8–12, 2025, 12:30 PM – 6:50 PM	Quarter	Quarter 3, Week 6

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
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I. CURRICULUM CONTENT, STANDARDS, AND LESSON COMPETENCIES

A. Content Standards	The learners demonstrate knowledge and understanding of: 1. systems of linear equations in two variables				
B. Performance Standards	By the end of the quarter, the learners are able to: 1. solve a system of linear equations graphically and algebraically				
C. Learning Competencies	By the end of the lesson, the learners: 1. define and illustrate a system of linear equations in two variables 2. solve a system of linear equations (with integer solutions) by graphing 3. classify the types of systems of linear equations based on the number of solutions				

	The learner should be able to: a. Recall the concepts	The learner should be able to: a. Outline and estimate	The learner should be able to: a. Solve and analyze the various	The learner should be able to: a. Cite and analyze the various components involved in classifying types of	The learner should be able to: a. Recall the concepts
--	--	---	--	---	--

ACT AS A
SOCRATIC
TUTOR

ACT AS AN
ASSESSMENT
DESIGNER



Republic of the Philippines
Department of Education
National Capital Region
SCHOOLS DIVISION OFFICE OF QUEZON CITY
MALIGAYA HIGH SCHOOL
Ilang Ilang St., Maligaya Park Subd.
Pasong Putik, Nova., Quezon City

Name: _____
Section: _____

Date: _____
Score: _____

Third Summative Test in Mathematics 8
S.Y. 2023–2024

Multiple Choice: *Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.*

1. A point can be represented as:
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
2. Which of the following best represents a plane?
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
3. A structure that consists of defined and undefined terms, axioms or postulates, and theorems is called:
A. Direct proof B. Indirect proof C. Law of Syllogism D. Mathematical system
4. Two lines that are not parallel ____ intersect.
A. always B. sometimes C. maybe D. never
5. Evaluate the following statement: "Perpendicular lines form right angles."

ACT AS A
STRUGGLING
STUDENT

ACT AS A
MATH OLYMPIAD
COACH

ACT AS AN
EXCEL/DESMOS
WIZARD

ACT AS A
MATHEMATICAL
HISTORIAN

PRINCIPLE #4

REFINE

What should
I change?

Specify
difficulty

Specify number
of items

Specify format

Specify
vocabulary

Specify length

Specify
constraints

Specify answer
format

“Add
VISUALS”

“Simplify the
LANGUAGE”

“Add a
STEP-BY-STEP
hint”

“Turn it into
MULTIPLE
CHOICE”

“Format for
LATEX”

“Make it a
SPOOKY
theme”

“Remove the
DECIMALS”

THINK
STEP-BY-STEP

“Explain your
reasoning before
THE ANSWER”

REFINE
UNTIL PERFECT

PRINCIPLE #5

SHOW
EXAMPLES

Can I show AI
what I want?

Show AI a
sample problem

Show AI
desired format

Show AI
desired solution

Show AI
desired tone

“Like this:

$$x^2 + 5x + 6 = 0”$$

“NOT like this:
 $x^2 + 100x = 0$ ”

“Use this tone:
A baker has x flour...”

“Show steps like:
Step 1: Simplify
Step 2: Solve”

“Include radicals:

$$2\sqrt{3} + 5\sqrt{3}”$$

“Format as a table:
| Input | Output |”

USE CASES

GENERATE
PROBLEMS

SAMPLE PROMPT: GENERATE PROBLEMS

“[Role] Act as a veteran Algebra teacher. [Context] My 9th-grade students are struggling with the distributive property, specifically when a negative sign is outside the parentheses. [Specific] Generate 8 practice problems. [Example] Include problems like $-3(x-4)$ but avoid simple ones like $2(x+5)$. [Refine] Format the output in a clean list with enough space for student work, and provide a separate answer key at the bottom.”

25. The figures are marked with their congruent parts. Determine the other congruent parts using the ASA congruence postulate.

A. $\overline{SR} \cong \overline{SR}$

B. $\overline{RQ} \cong \overline{RE}$

C. $\angle SRQ \cong \angle SRE$

D. $\angle RSQ \cong \angle RSE$

26. To prove that $\triangle QRW \cong \triangle VWR$, which postulate can be used?

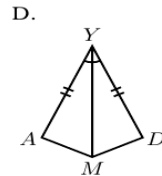
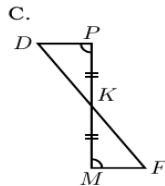
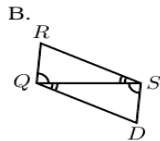
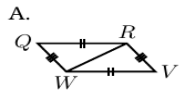
A. ASA postulate

B. SAS postulate

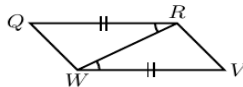
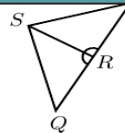
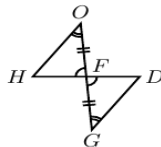
C. SSS postulate

D. AAS postulate

27. Which of the following pairs of triangles are congruent by the SAS postulate?



28. Which postulate can be used to conclude that $\triangle HOF \cong \triangle DGF$?



CREATE ANSWER KEYS

SAMPLE PROMPT: ANSWER KEYS

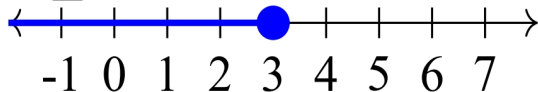
“[Role] Act as a meticulous Math TA. [Context] I am grading a quiz on the Quadratic Formula. [Specific] Create a detailed answer key for the equation $2x^2 - 7x + 3 = 0$. [Example] Show the steps exactly like this: Step 1: Identify a, b, c; Step 2: Substitution; Step 3: Simplification. [Refine] Use LaTeX for all mathematical notation and highlight the two final roots in bold.”

Activity 3.4: Graphing Solutions of Linear Inequalities on a Number Line

Total points = 38

1. $x + 2 - 2 \leq 5 - 2$ ✓

$x \leq 3$ ✓



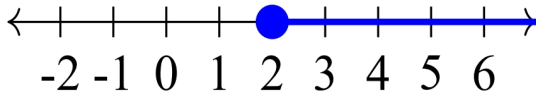
✓

2. $3x - 4 + 4 \geq 7 + 4$ ✓

$\frac{3x}{3} \geq \frac{11}{3}$ ✓

$\frac{6x}{6} \geq \frac{12}{6}$ ✓

$x \geq 2$ ✓



✓

7. $3x - 2x + 5 - 5 \leq 2x - 2x + 9 - 5$

✓

PREDICT MISTAKES

SAMPLE PROMPT: PREDICT MISTAKES

“[Role] Act as an expert Math Pedagogy Researcher. [Context] I am teaching 'Subtracting Polynomials' to 8th graders tomorrow. [Specific] List the 3 most common conceptual errors students make with this topic. [Example] Specifically focus on the 'integer sign error' when distributing the negative. [Refine] Present this as a table: Error Name | Example of the Mistake | How to explain the fix.”

WRITE WORD PROBLEMS

SAMPLE PROMPT: WORD PROBLEMS

“[Role] Act as a creative curriculum designer. [Context] My students are 14-year-olds who love sports and video games, and we are studying Pythagorean Theorem. [Specific] Write 4 word problems that require finding the hypotenuse. [Example] Use a scenario like a player running a diagonal route in basketball. [Refine] Avoid overly complex vocabulary and ensure every final answer is a terminating decimal or a whole number.”

Practice Exercises 3.14

Solve the following word problems using systems of linear equations.

1. A ticket for a movie costs ₱150 for adults and ₱100 for children. If 20 tickets were sold for a total of ₱2,500, how many adult and child tickets were sold?
2. The sum of two numbers is 40. Their difference is 8. What are the numbers?
3. One number is twice another. Their sum is 36. Find the numbers.
4. A shop sells pens for ₱12 each and pencils for ₱8 each. If 15 items are sold for a total of ₱148, how many pens and pencils were sold?
5. The perimeter of a rectangle is 34 cm. The length is 3 cm more than the width. Find the dimensions of the rectangle.

EXPLAIN
STEP-BY-STEP

SAMPLE PROMPT: STEP-BY-STEP

“[Role] Act as a supportive math tutor. [Context] I am helping a student who understands basic equations but is stuck on solving variables on both sides. [Specific] Break down the solution for $5x - 3 = 2x + 12$. [Example] Use the 'balancing the scales' analogy for each step. [Refine] Number the steps clearly and include a 'Check Your Work' section at the end where the student substitutes the answer back into the original equation.”

DESIGN ASSESSMENTS

SAMPLE PROMPT: DESIGN ASSESSMENTS

“[Role] Act as a formal assessment coordinator. [Context] I need a mid-unit quiz for a Grade 10 Trigonometry unit. [Specific] Create a 5-question quiz covering the Sine, Cosine, and Tangent ratios. [Example] Include one 'finding a side' problem, one 'finding an angle' problem, and one multi-step word problem involving a ladder against a wall. [Refine] Organize the quiz by difficulty from easiest to hardest and include a rubric for the word problem.”

Quiz 3.6: Proving the Congruence of Triangles

Multiple Choice: Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.

1. Which theorem states that if the legs of one right triangle are congruent to the legs of another right triangle, then the triangles are congruent?

A. HA Congruence Theorem

C. LA Congruence Theorem

B. HL Congruence Theorem

D. LL Congruence Theorem

2. "If two angles and a non-included side of one triangle are congruent to the corresponding two angles and a non-included side of another triangle, then the triangles are congruent." This is stated in:

A. AAS Congruence Theorem

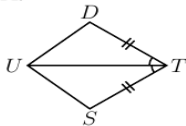
C. HL Congruence Theorem

B. LL Congruence Theorem

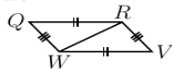
D. LA Congruence Theorem

3. Which of the following pairs of triangles are congruent and can be proved by ASA Postulate?

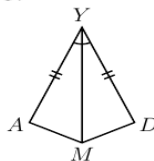
A.



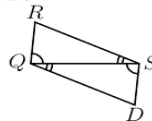
B.



C.



D.



4. Which of the following pairs of triangles are congruent and can be proved by HL Theorem?

SCAFFOLD LEARNING

SAMPLE PROMPT: SCAFFOLD LEARNING

“[Role] Act as a differentiation specialist. [Context] I am teaching 'Area of Composite Figures' to a diverse class. [Specific] Take a single L-shaped polygon and create three versions of the task. [Example] Version A: All side lengths given; Version B: One side length missing (requires subtraction); Version C: A 'reverse' problem where area is given and a side is missing. [Refine] Write clear instructions for each version and include a 'Help Box' with a hint about dividing the shape into rectangles.”

CREATE
FAKE MISTAKES

SAMPLE PROMPT: CREATE FAKE MISTAKES

“[Role] Act as a student who is prone to ‘careless errors.’ [Context] We are practicing the distributive property with negatives. [Specific] Solve the problem $-4(3x - 5)$ incorrectly. [Example] Make the specific error of forgetting to flip the sign of the second term (e.g., $-12x - 20$). [Refine] Write out the ‘student’s’ work line-by-line and add a prompt for my real students: ‘Identify the mistake and explain how to fix it.’”

FIND APPLICATIONS

SAMPLE PROMPT: FIND APPLICATIONS

“[Role] Act as a STEM Career Counselor. [Context] I am introducing 'Logarithms' to students who think it is just abstract button-pressing. [Specific] Provide 3 modern, real-world applications of logarithms. [Example] Include one related to tech (like data science) and one related to music or sound. [Refine] For each application, write a 2-sentence explanation that a teenager would find interesting, and provide one 'hook' question to start a class discussion.”

DESIGN RUBRICS

SAMPLE PROMPT: DESIGN RUBRICS

“[Role] Act as a Lead Evaluator for Math Department. [Context] I am assessing a project where students must model a real-world scenario using 'Systems of Linear Inequalities.' [Specific] Create a 4-point rubric across three categories: Mathematical Accuracy, Graphing Precision, and Written Justification. [Example] For a '4' in Justification, the student must explain why a point in the shaded region satisfies both constraints. [Refine] Format this as a table for easy copy-pasting into a document.”

PROMPT MAKEOVER

Make a lesson
about fractions.

Make difficult
math problems.

Explain
algebra.

Make a quiz.

Create
PowerPoint slides
for Grade 8.

TIPS

START
SIMPLE

BE
CONVERSATIONAL

ALWAYS REVIEW

AI makes
MISTAKES

YOU'RE THE
EXPERT

ASK FOR
LATEX

L^AT_EX

“Format these
equations for
OVERLEAF”



overleaf.com/project/662213bf39b170ac8c7fa8be



File Edit Insert View Format Help

Hello world ▾



Share

Upgrade



main.tex



main.tex

```
1 \documentclass{article}
2 \usepackage{graphicx} % Required for inserting
  images
3
4 \title{Hello world}
5 \author{cityofsmiles }
6 \date{April 2024}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
```

Editing ▾

Recompile ▾



43% ▾



Hello world

cityofsmiles

April 2024

1 Introduction

hello world

WHAT NOW?

EXPERIMENT



📦 Upgrade



🔔 Memory off ⓘ



Create better and faster images >

Edit your photos, make images of yourself, and try new styles



Ask ChatGPT



PRACTICE

A close-up of Baby Yoda from Star Wars: The Mandalorian. He is sitting in a wooden crate, wrapped in a light-colored cloth. He has a serious expression on his face. The background is blurred, showing some outdoor structures.

**DID YOU PRACTICE
TODAY?**

PRINCIPLE #1

BE
SPECIFIC

PRINCIPLE #2

GIVE
CONTEXT

PRINCIPLE #3

ASSIGN
A ROLE

PRINCIPLE #4

REFINE

PRINCIPLE #5

SHOW
EXAMPLES

SPECIFIC
CONTEXT
ASSIGN A ROLE
REFINE
EXAMPLES

SCARE

IMPROVE

TRANSFORM
YOUR
TEACHING

CHALLENGE

Choose one lesson
you will teach in
the second term.

Using SCARE, create 3:

1. Worksheet

2. Assessment

3. Answer key

4. Slides

5. Handout

6. Study notes

7. Flashcards

Don't be SCAREd of
AI. Use SCARE to
master it.

QUESTIONS?

THANK YOU

FOR YOUR ATTENTION

